

Data Table

	Forces	Resultant - $\vec{R}$ (magnitude and direction angle)	
		Experimental	Analytical
Vector Addition 1	$F_1 = 0.125 \text{ g}\cdot\text{kg}$ and $\theta_1 = 30^\circ$ $F_2 = 0.125 \text{ g}\cdot\text{kg}$ and $\theta_2 = 120^\circ$	$R =$ $\theta_R =$	$R =$ $\theta_R =$
Vector Addition 2	$F_1 = 0.075 \text{ g}\cdot\text{kg}$ and $\theta_1 = 20^\circ$ $F_2 = 0.125 \text{ g}\cdot\text{kg}$ and $\theta_2 = 120^\circ$	$R =$ $\theta_R =$	$R =$ $\theta_R =$
Vector Addition 3	$F_1 = 0.125 \text{ g}\cdot\text{kg}$ and $\theta_1 = 30^\circ$ $F_2 = 0.175 \text{ g}\cdot\text{kg}$ and $\theta_2 = 80^\circ$ $F_3 = 0.165 \text{ g}\cdot\text{kg}$ and $\theta_2 = 150^\circ$	$R =$ $\theta_R =$	$R =$ $\theta_R =$
Vector Resolution 1	$F = 0.125 \text{ g}\cdot\text{kg}$ and $\theta = 60^\circ$	$F_x =$ $F_y =$	$F_x =$ $F_y =$
Vector Resolution 2	$F = 0.155 \text{ g}\cdot\text{kg}$ and $\theta = 125^\circ$	$F_x =$ $F_y =$	$F_x =$ $F_y =$

Show your work. Use the back of this sheet or a separate sheet.

Questions

Question 1 Two vectors $\vec{A}$ and $\vec{B}$ have magnitudes A and B , respectively. What are the largest and smallest values of the magnitude of $\vec{A} + \vec{B}$, the sum of the two vectors? Give your answer in terms of A and B . Remember that the magnitude of a vector must be positive.

Question 2 Give a different derivation of the analytical results of Vector Addition 1, using the parallelogram rule and simple geometry. Note that since the two vectors have the same magnitude and are perpendicular, the parallelogram is a square.

Question 3 Using the values given in Vector Addition 2, find the magnitude and direction angle of the difference of the two vectors, $\vec{F}_2 - \vec{F}_1$.

Question 4 When experimentally determining the components of $\vec{F}$ in Vector Resolution 2, at what angles were the pulleys placed? Give all three angles used.